Tables

1 A player is rolling two dice and calculating their sum. They draw a table of all the possible dice rolls for two dice and what they sum to:

Find the probability the dice will sum to 8.

	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

2 A player is rolling two dice and calculating their difference, that is the largest number minus the smaller number. They draw a table of all the possible dice rolls for two dice and what their difference is:

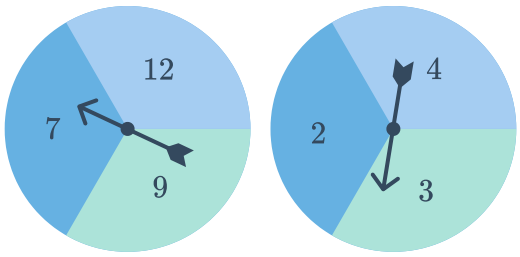
a List the sample space for the difference of two dice.

b What is the probability the dice will have a difference of 0?

	1	2	3	4	5	6
1	0	1	2	3	4	5
2	1	0	1	2	3	4
3	2	1	0	1	2	3
4	3	2	1	0	1	2
5	4	3	2	1	0	1
6	5	4	3	2	1	0

3 The following two spinners are spun and the sum of their respective spins are recorded:

a Complete the following table to represent all possible combinations:



Spinner	2	3	4
7		10	
9	11		13
12	14		16

b Find the probability that:

i The first spinner lands on an even number and the sum is even.

ii The first spinner lands on a prime number and the sum is odd.

iii The sum is a multiple of 3.

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4 A fair die is rolled twice.

a Create a table displaying all possible outcomes.

b Find the probability of rolling a:

i 3 and a 6 in any order.

ii 3 and then a 6.

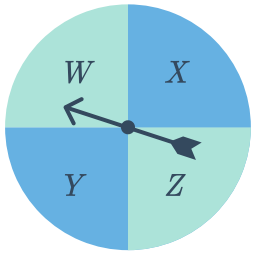
iii a double

iv an odd and an even number

5 The following spinner is spun and a normal six-sided die is rolled. The result of each is recorded:

a Complete the following table to represent all possible combinations:





	W	X	Y	Z
1	1, W	1, ⬚	1, Y	1, Z
2	⬚, W	2, X	2, Y	2, Z
3	3, W	3, X	3, Y	⬚, ⬚
4	4, W	4, X	4, Y	4, Z
5	5, W	5, X	⬚, ⬚	5, Z
6	6, W	6, X	6, Y	6, Z

b State the total number of possible outcomes.

c Find the probability that:

i The spinner lands on X and the dice rolls a prime number.

ii The spinner lands on W and the dice rolls a factor of 6.

iii The spinner doesn't land on Z or the dice doesn't roll a multiple of 3.

6 The following spinner is spun and a normal six-sided die is rolled. The product of their respective results is recorded.

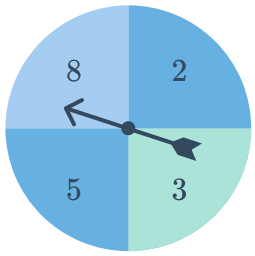
a Construct a table to represent all possible outcomes.

b Find the probability of an odd product.

c Find the probability of rolling a 5 on the dice and scoring an even product.

d Find the probability of spinning a 3 on the spinner or scoring a product which is a multiple of 4.





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7 Xavier is choosing an outfit for the day and has 3 shirts (cyan, pink, and white) and 4 ties (black, grey, red, and yellow) to select from.

a Complete the following table to show all the possible outfits Xavier could wear:

	Cyan (C)	Pink (P)	White (W)
Black (B)	C, B	$\square, B$	W, B
Grey (G)	C, G	P, G	$W, \square$
Red (R)	$\square, \square$	P, R	W, R
Yellow (Y)	C, Y	$\square, \square$	W, Y

b How many different outfits are possible?

c What is the probability that he wears a black tie?

d What is the probability that he wears a white shirt?

8 200 people were questioned about whether they voted for Labor, Liberal or Greens last election and who they'll vote for this election.

a Fill in the missing values in the table of results:

	Labor (next election)	Liberal (next election)	Greens (next election)	Total
Labor (last election)	25	41		70
Liberal (last election)	16		1	90
Greens (last election)	9	26		
Total		140	10	200

b Determine the probability that a randomly selected person will vote Labor next election given that they voted Liberal last election.

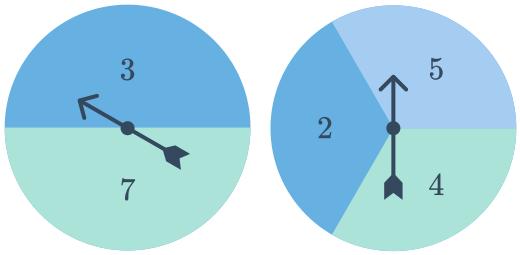
c Determine the probability that a randomly selected person did not vote Green last election given that they will vote Green this election.

d If a person voted Labor or Liberal last election, what is the probability they'll vote Liberal this election?

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9 The following two spinners are spun and the sum of their respective spins is recorded:

a Complete the following table to represent all possible outcomes:



Spinner	2	4	5
3			
7			

b Find the probability that:

i The sum is less than 13.

ii The sum is odd.

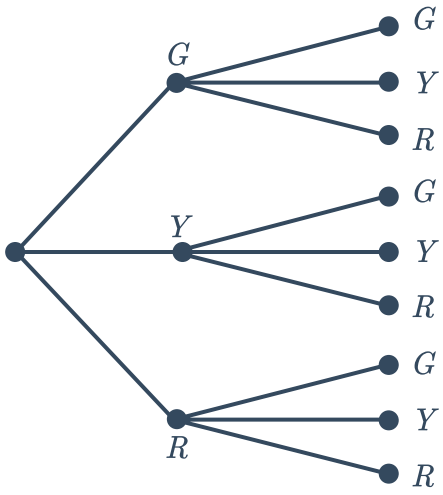
iii A 7 was spun given that the sum is less than 12.

iv The sum is odd given that a 4 was spun.

Tree diagrams

10 On the island of Timbuktoo, the probability that a set of traffic lights shows red, yellow or green is equally likely. Christa is travelling down a road where there are two sets of traffic lights.

What is the probability that both sets of traffic lights will be red?



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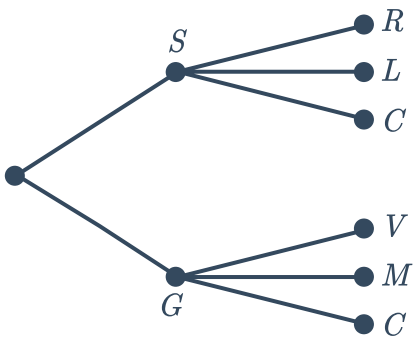
11 An ice-cream shop offers one flavour of ice-cream at a discounted price each day. There are six flavours for the owner to choose from, that could be discounted. Three of the flavours are sorbet and the other three are gelato.

The three sorbet flavours are raspberry (R), lemon (L) and chocolate (C). The three gelato flavours are vanilla (V), mint (M) and chocolate (C). Each decision has equal probability.

a List the sample space for which ice-cream type and flavour could be chosen.

b Find the probability that the lemon sorbet is selected.

c Find the probability that the flavour will be chocolate.



12 Hadyn is looking for a way to randomly choose a prime number between 1 and 20.

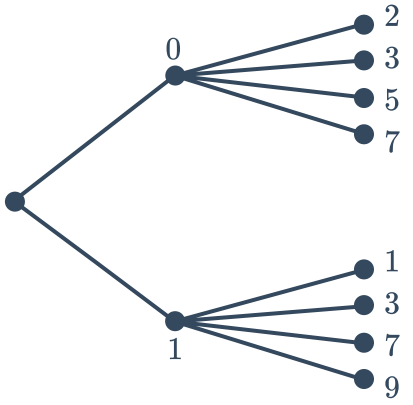
He first flips a coin to decide 0 or 1 and depending on the result he then rolls one of two four-sided dice, shown in the tree diagram. He arrives at his number by putting the coin number in front of the dice number.

a List the sample space of numbers created.

b Find the probability that:

i The number 19 will be the result.

ii The number will end in a 3.



13 Three cards labelled 2, 3 and 4 are placed face down on a table. Two of the cards are selected randomly to form a two-digit number. The outcomes are displayed in the following probability tree diagram:

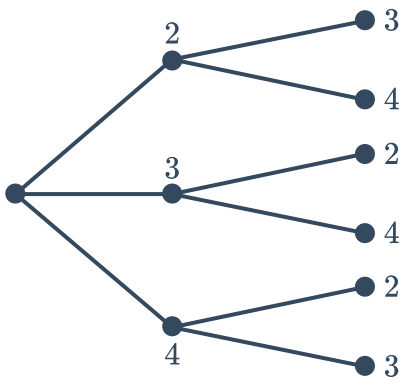
a List the sample space of two digit numbers produced by this process.

b Find the probability that:

i 2 appears as a digit in the number.

ii The sum of the two selected cards is even.

iii The number formed is greater than 40.



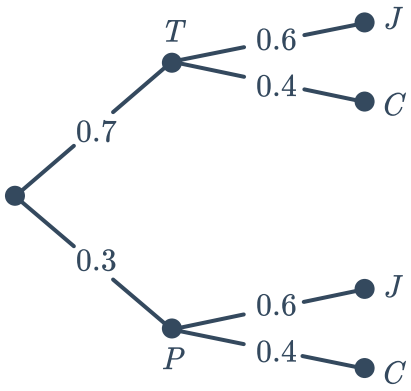
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14 For breakfast, Maria has something to eat and drinks a hot drink. She will eat either toast (T) or porridge (P) and will drink either juice (J) or coffee (C).

The chance of Maria making toast is 0.7. The chance of Maria drinking coffee is 0.4.

a Find the probability that Maria drinks juice and eats toast.

b Find the probability that Maria drinks juice or eats toast.



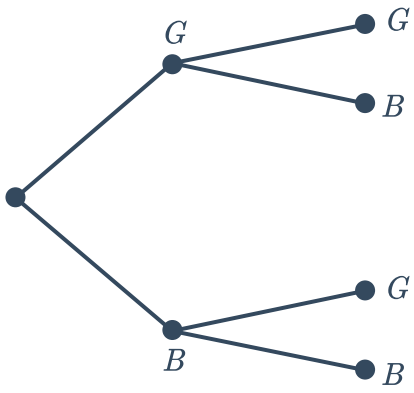
15 Han owns four green ties and three blue ties. He selects one of the ties at random for himself and then another tie at random for his friend.

a Write the probabilities for the outcomes on the edges of the probability tree diagram:

b Calculate the probability that:

i Han selects a blue tie for himself.

ii Han selects two green ties.



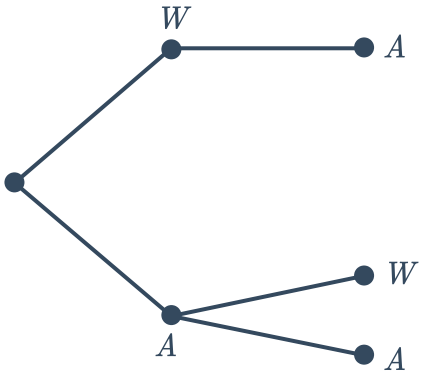
16 Bart is purchasing a plane ticket to Adelaide. He notices there are only 4 seats remaining, 1 of them is a window seat (W) and the other 3 are aisle seats (A). His friend gets on the computer and purchases a ticket immediately after. The seats are randomly allocated at the time of purchase.

a Write the probabilities for the outcomes on the edges of the probability tree diagram for the seat Bart receives and the seat his friend receives:

b Find the probability that:

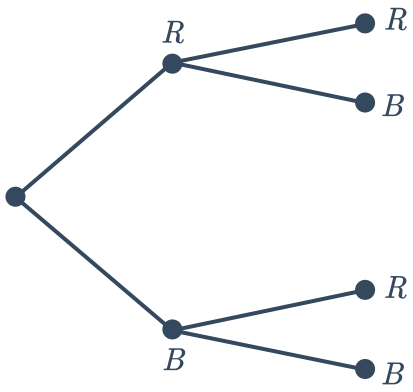
i Bart's friend has an aisle seat.

ii Bart's friend receives an aisle seat if Bart has a window seat.



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17 Sally is drawing 2 cards from a deck of 52 cards. She draws the first card and checks whether it is red (R) or black (B). Without replacing her first card, she draws the second card and records its colour.



a Write the probabilities for the outcomes on the edges of the probability tree diagram:

b What is the probability that Sally draws a black card and then a red card?

c State whether each of the following events has an equal probability to drawing a black then a red card:

i Drawing a red card and then a black card.

ii Drawing a red card and then another red card.

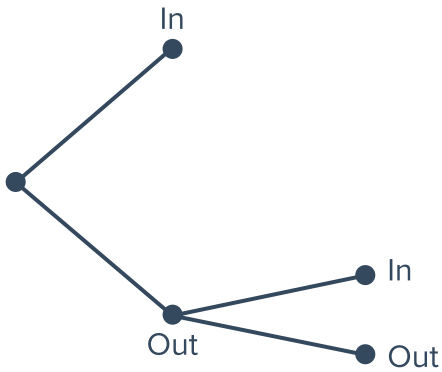
iii Drawing at least one black card.

iv Drawing a black card and then another black card.

v Drawing one black card and one red card in any order.

18 In tennis, if a player's first serve goes out, the player takes a second serve. If it goes in, they don't need to take a second serve. A player serves with the following probabilities:

- First serve goes in - 0.55
- Second serve goes in - 0.81



a Write the probabilities for the outcomes on the edges of the probability tree diagram:

b Find the probability that:

i The player needs to make a second serve.

ii Both of the player's serves go out.

19 A coin is tossed twice.

a Construct a tree diagram showing the results of the given experiment.

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b Use the tree diagram to find the probability of tossing:

i Exactly 1 head.

ii 2 heads.

iii No heads.

iv 1 head and 1 tail.

20 Han watches two episodes of TV each night. He begins with either News or Current Affairs. If he watches Current Affairs, he might watch either Comedy, Horror or Animation next. If he watches News, he will always choose a Comedy to watch afterwards.

a Construct a tree diagram of all possible options.

b If, at every stage, the possible outcomes of each choice are equally likely, what is the probability Han watches a Comedy?

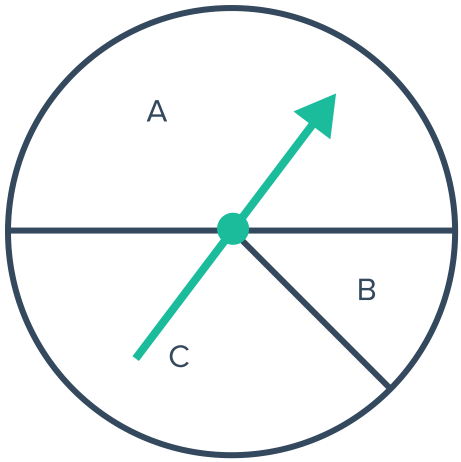
21 A coin is tossed, then the spinner shown is spun and either lands on A, B or C:

a Construct a probability tree diagram that correctly shows all possible outcomes and probabilities.

b Find the probability of:

i Landing on tails and the spinner landing on C.

ii Landing on tails, or the spinner landing on C, or both.



22 Five ordinary dice are rolled. What is the probability that the results are all sixes?

23 Neville has found that when playing chess against the computer, he wins $\frac{1}{3}$ of the time.

a Find the probability that:

i He wins two games in a row.

ii He wins three games in a row.

iii He wins at least one of three games.

b Calculate whether Neville has a better chance at winning at least one of two games or winning at least one of three games.

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24 Mr and Mrs Smith are starting a family. When having each child, they assume that having a girl is just as likely as having a boy.

- **a** Find the probability that:
  - **i** The first child is a girl.
  - **ii** The first child is a girl and the second child is a boy.
  - **iii** The first two children are girls and the third child is a boy.
- **b** The Smiths have three children, all girls. What is the probability that the next child will be a boy?
- **c** What is the probability that in a four-child family, the first three are girls and the fourth is a boy?

25 Consider tossing a normal fair coin:

- **a** If you have already tossed the coin 10 times, what is the chance that on the next toss it will land on tails?
- **b** If I've tossed it 5 times, what is the chance the first three were a tail and the last two were a head?

2D Grids

26 Two coins are tossed at the same time.

- **a** Display the sample space in a 2D grid.
- **b** How many different outcomes are possible?
- **c** Find the probability of tossing a heads and a tail.

27 A coin and a six-sided die are thrown at the same time.

- **a** Display the sample space in a 2D grid.
- **b** How many different outcomes are possible?

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28 Ben has 3 shirts, each in a different colour: crimson (C), pink (P) and white (W), and 4 ties, each in a different colour: blue (B), grey (G), red (R) and yellow (Y).

a Complete the table to show the sample space of the possible combinations Ben could wear.

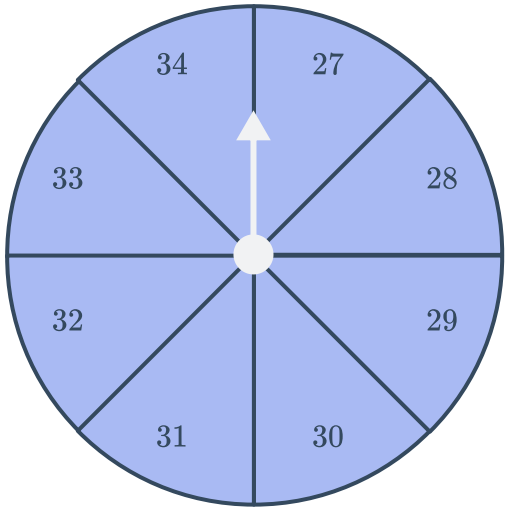
b Construct a 2D grid to show the sample space of the possible combinations Ben could wear.

c How many different combinations are possible?

	C	P	
B	C,B	P,B	
G			G,W
R		P,R	W,R
	C,Y	P,Y	

29 The following spinner is spun and a six-sided die is rolled at the same time:





a Construct a 2D grid to display the sample space.

b How many different outcomes are possible?

c Find the probability of getting two even numbers.

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